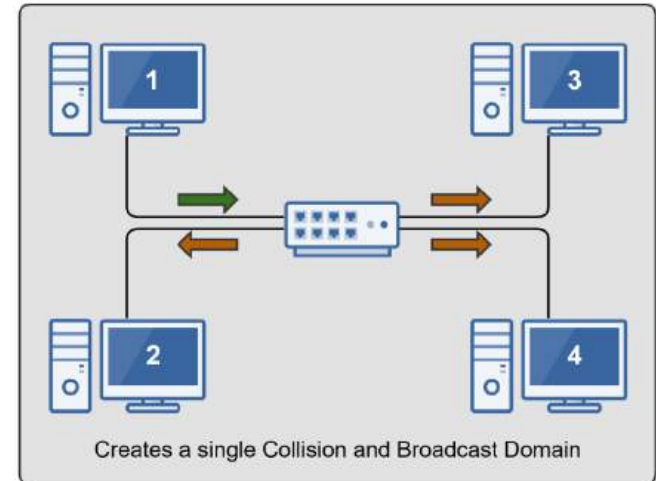


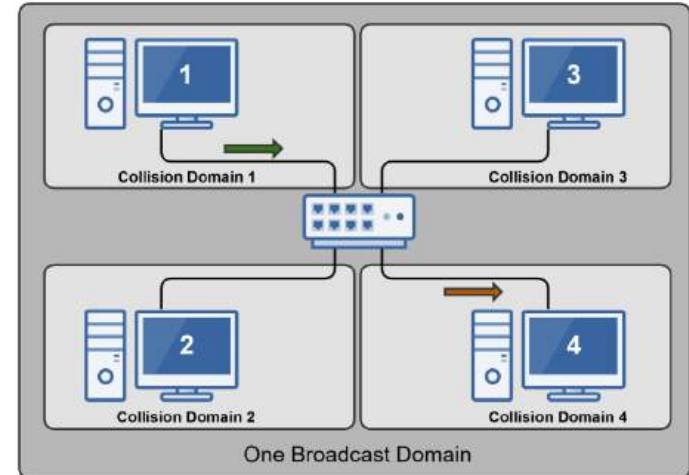
Hubs

- Used to Connect Devices Together Within a Network
- Used in Early Networks; Replaced by Switches
- “Multi-Port Repeater”
 - Traffic goes in one port and is repeated (broadcasted) out every other port
 - OSI Layer 1 Device
 - Dumb Network Device
 - Causes increased network collision errors
- Much Less Efficient than a Switch
- Legacy Equipment No Longer Used



Switches

- Connects Devices Together Just Like a Hub
- Intelligent Network Device (OSI Layer 2)
- Memorizes the **MAC Address** of Each Device Connected to It via a **MAC Address Table**, sometimes called a **Content Addressable Memory (CAM) Table**
- Pays attention to *Source* and *Destination* **MAC addresses** during Communication Process
- Use Application-Specific Integrated Circuitry (**ASIC**), which makes them Extremely Fast
- Breaks up Collision Domains
 - Traffic Goes in One Port and Is Repeated out to Only Destination Port
 - Designed for High Bandwidth
 - Standard in Today's Network Infrastructure



1-Unmanageable Switch



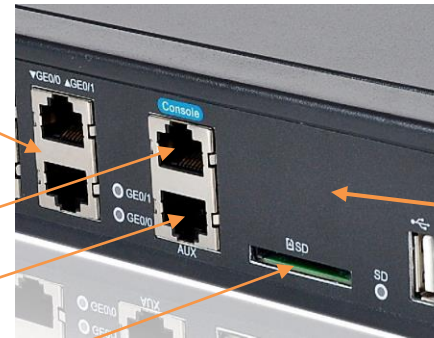
2-Manageable Switch

2 Up Link

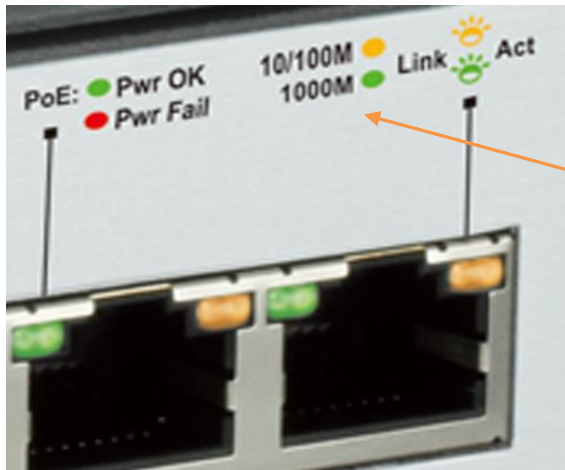
Console

AUX

SD Card

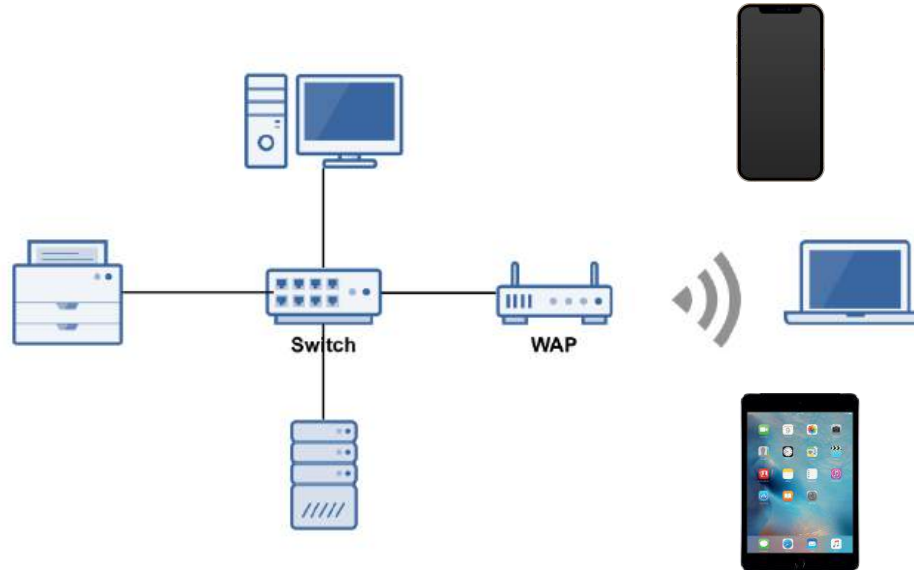


3-PoE Switch



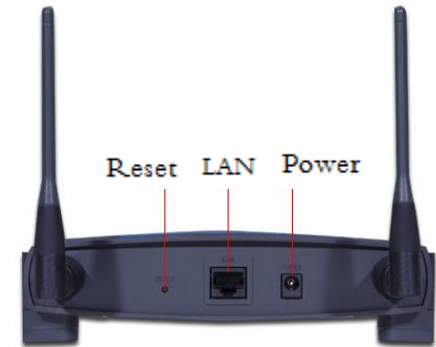
Wireless Access Point (WAP)

- A wireless access point (WAP) is a bridge that extends the wired network to the wireless network.
- Just like a switch, it's a Data Link Layer 2 device.
- **Note:** A WAP is not a router.



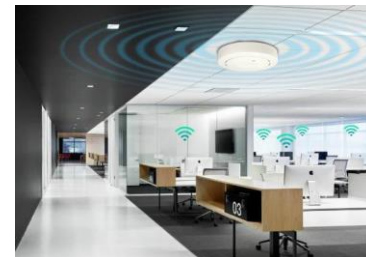
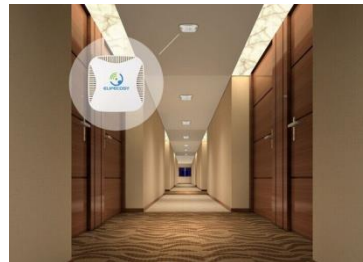


Front



Rear

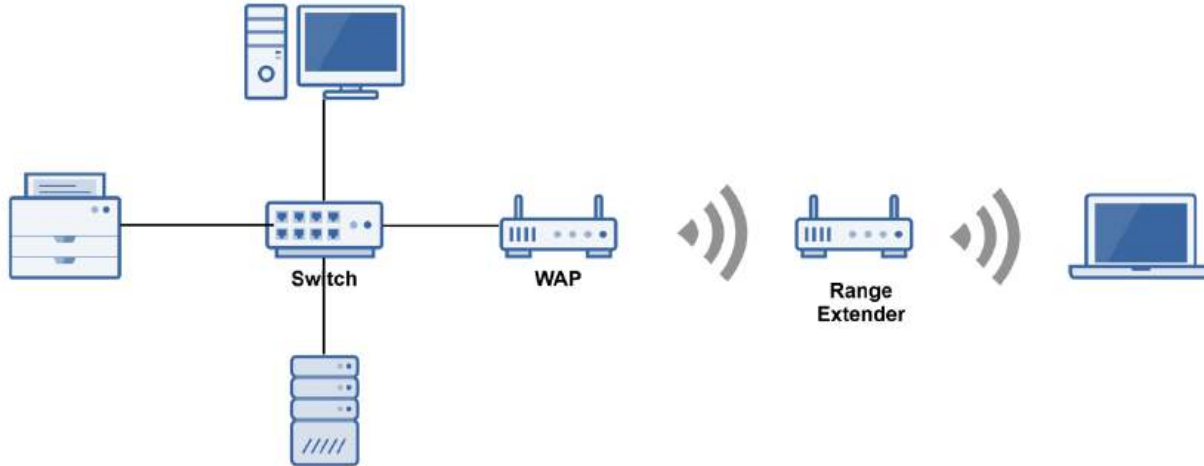
Ceiling Access Point



Ceiling Access Point

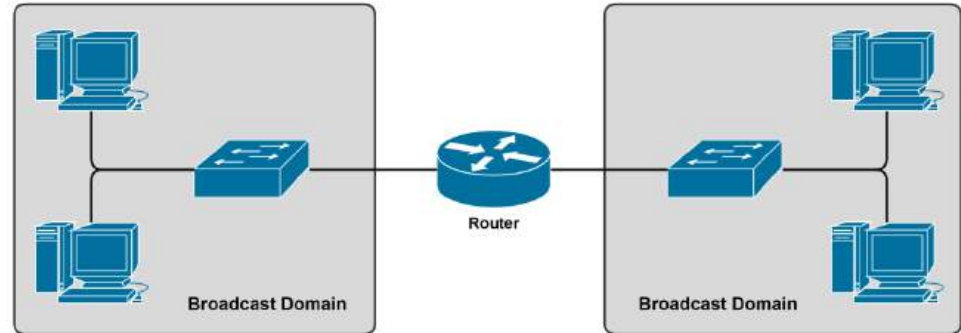
Wireless Ranger Extender

- Extends the range of a wireless network by acting as a wireless repeater.
- Rebroadcasts radio frequencies from the wireless network it is associated with



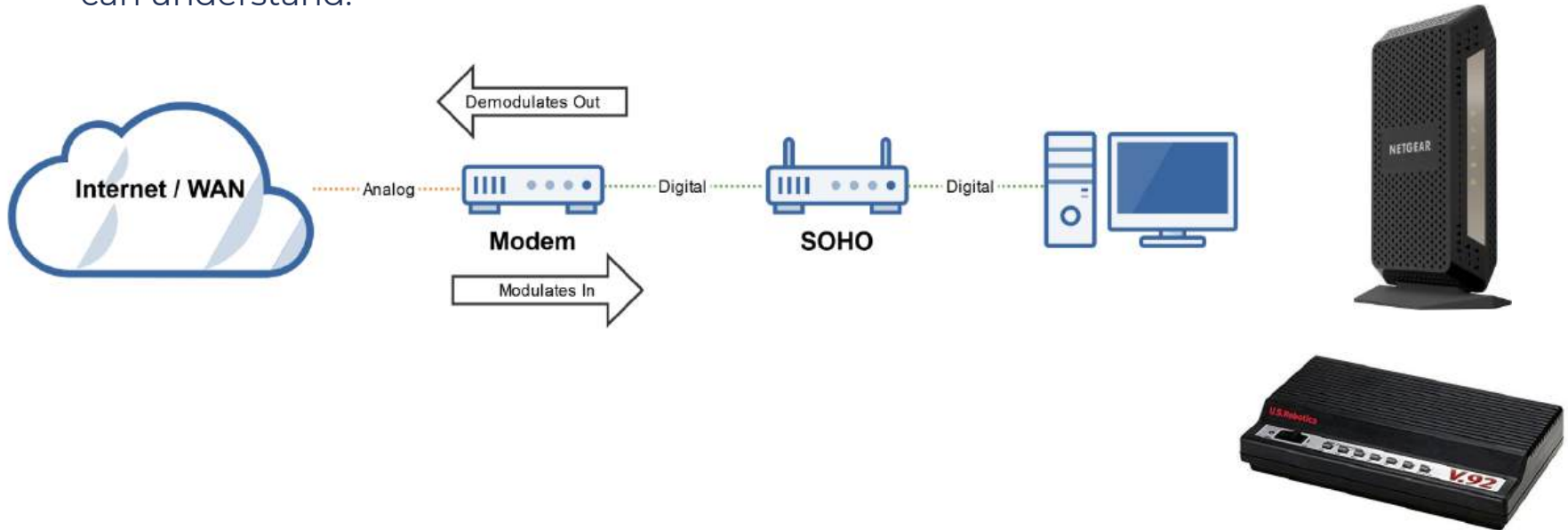
Routers

- Used to Connect Different Networks Together
- Routes Traffic Between Networks using **IP Addresses**
- Uses Intelligent Decisions (Routing Protocols) to Find the Best Way to Get a Packet of Information from One Network to Another.
- Break Up Broadcast Domains
- **OSI Layer 3 Device**
 - Layer 3 = Router
 - Layer 2 = Switch
 - Layer 1 = Hub



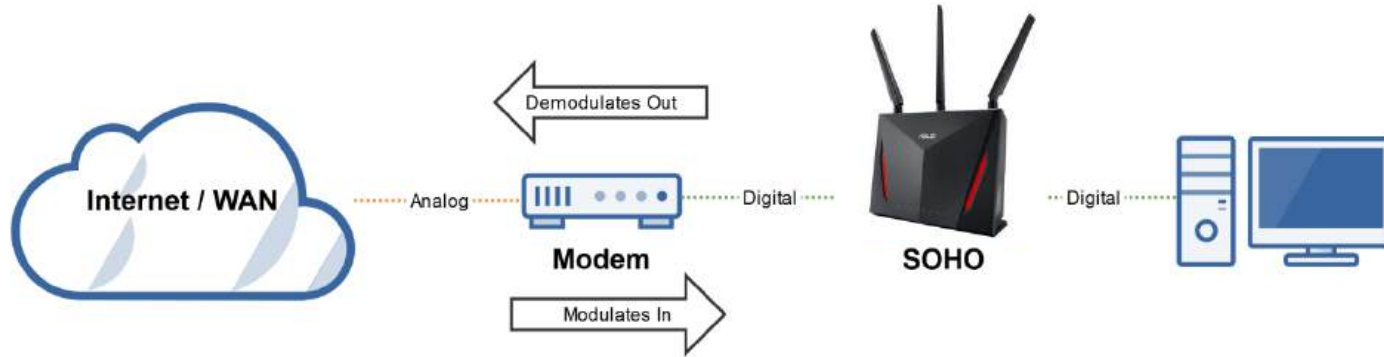
Modems (Modulators/Demodulators)

- Modems modulate one signal to another, such as analog to digital.
- For example, modulating a telephone analog signal into a digital signal that a router can understand.



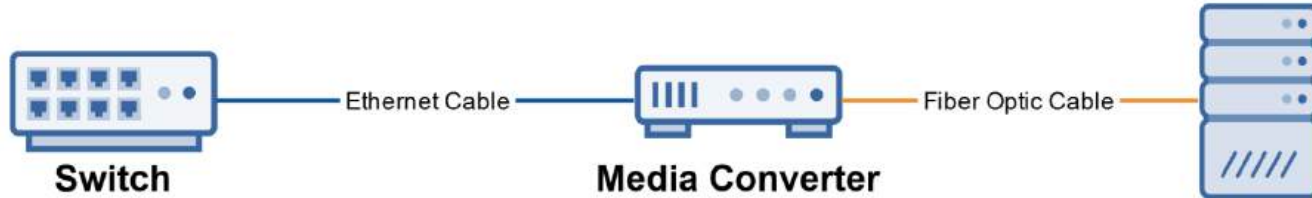
Small Office Home Office (SOHO) Device

- All-In-One Wireless Router with Expanded Capabilities:
 - Router, Wireless Access Point, Firewall, Switch, DHCP Server, NAT Device, File Server, etc.



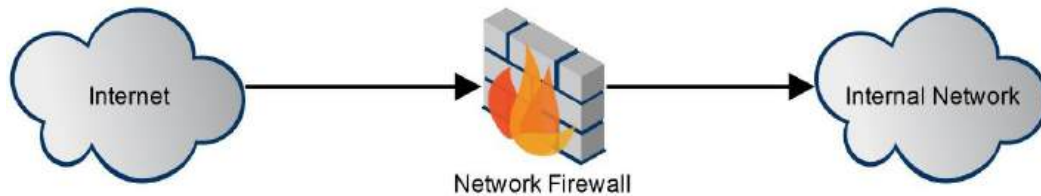
Media Converters

- Like its name implies, it converts one media type to another.
- **Layer 1 Device:** Performs physical layer signal conversion.
- Ethernet to fiber optic media converters are commonly used.



Firewalls

- Firewalls are the foundation of a defense-in-depth network security strategy.
- They protect your network from malicious activity on the Internet.
- Prevent unwanted network traffic on different networks from accessing your network.
- Firewalls do this by filtering data packets that go through them.
- They can be a standalone network device or software on a computer system, meaning **network-based (hardware)** or **host-based (software)**.



Types of Firewalls

Packet Filtering Firewalls

- 1st Generation & Most Basic
- Basic Filtering Rules

Circuit-Level Firewalls

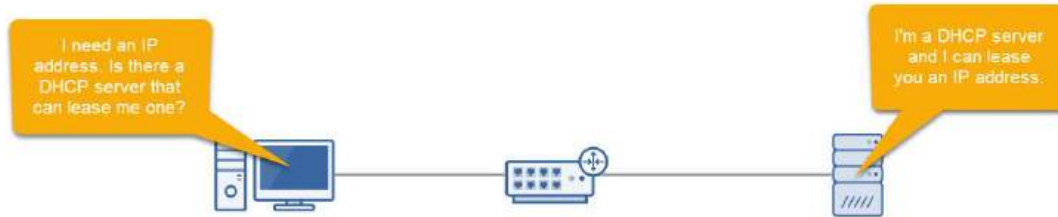
- 2nd Generation
- Monitors Valid/Invalid TCP Sessions

Application Layer 7 (NGFW) Firewalls

- 3rd Generation
- Much more Advanced; Covered Later in Course

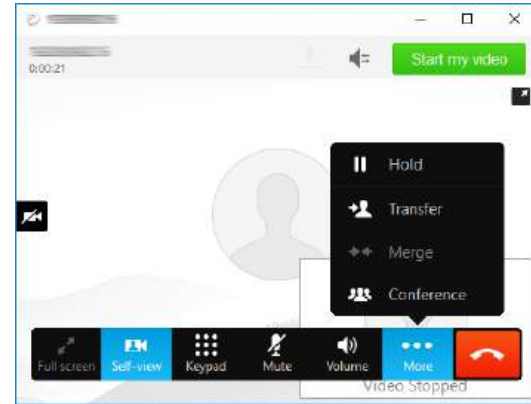
DHCP Server

- Dynamic Host Configuration Protocol (DHCP) Server
- Automatically Assigns IP Addresses to Hosts
- Makes Administering a Network Much Easier
- An Alternative is Static IP addressing
- We'll Talk More About DHCP Later in the Course



Voice over IP (VoIP) Endpoints

- Most phone systems run over IP networks via dedicated protocols, such as the Session Initiation Protocol (SIP), both in-home and office environments.
- VoIP endpoint devices are hardware devices (phones) or software, such as Cisco Jabber, that allow you to make phone calls.



Rack

Wall Rack



Ground Rack

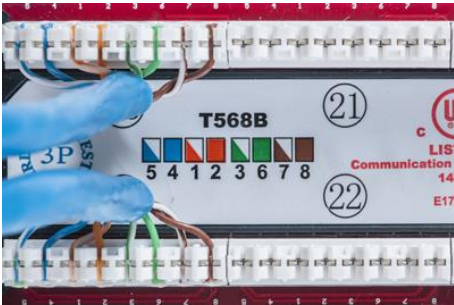


Wallmount 8U

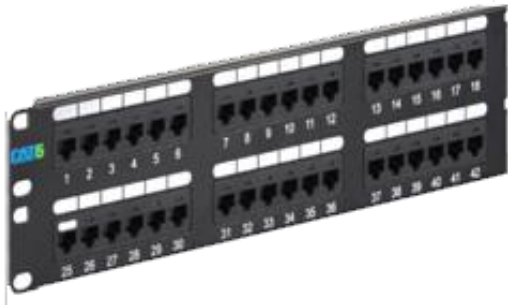


Patch Panel

Patch Panel (Ethernet Cable)



Rear



Front

Modular



Modular



Cable Manager



Cable Manager



Patch Cords

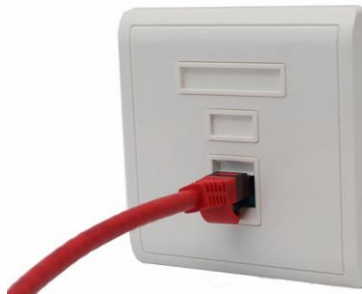


Patch Cord UTP

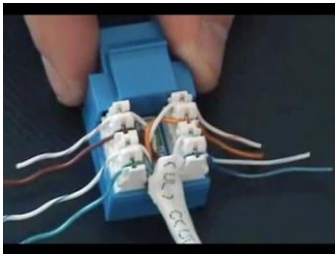


Patch Cord Fiber

Face Plate



Node (Modular Jack)



Ethernet Coupler

